

Introduction to User Modeling

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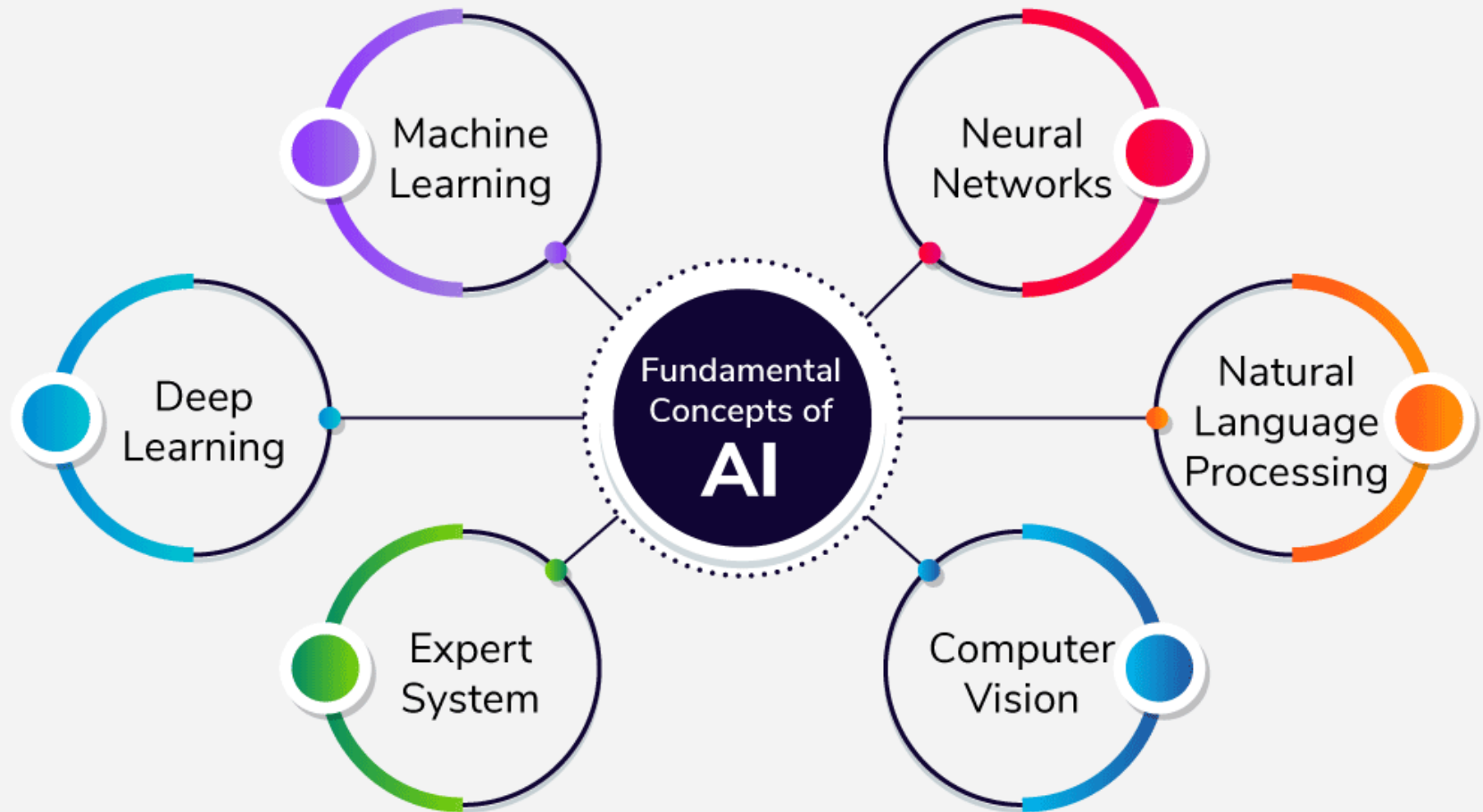
Comp 4957

Technical Programming

Computer Science Technology

BCIT

Fundamental Concepts of AI

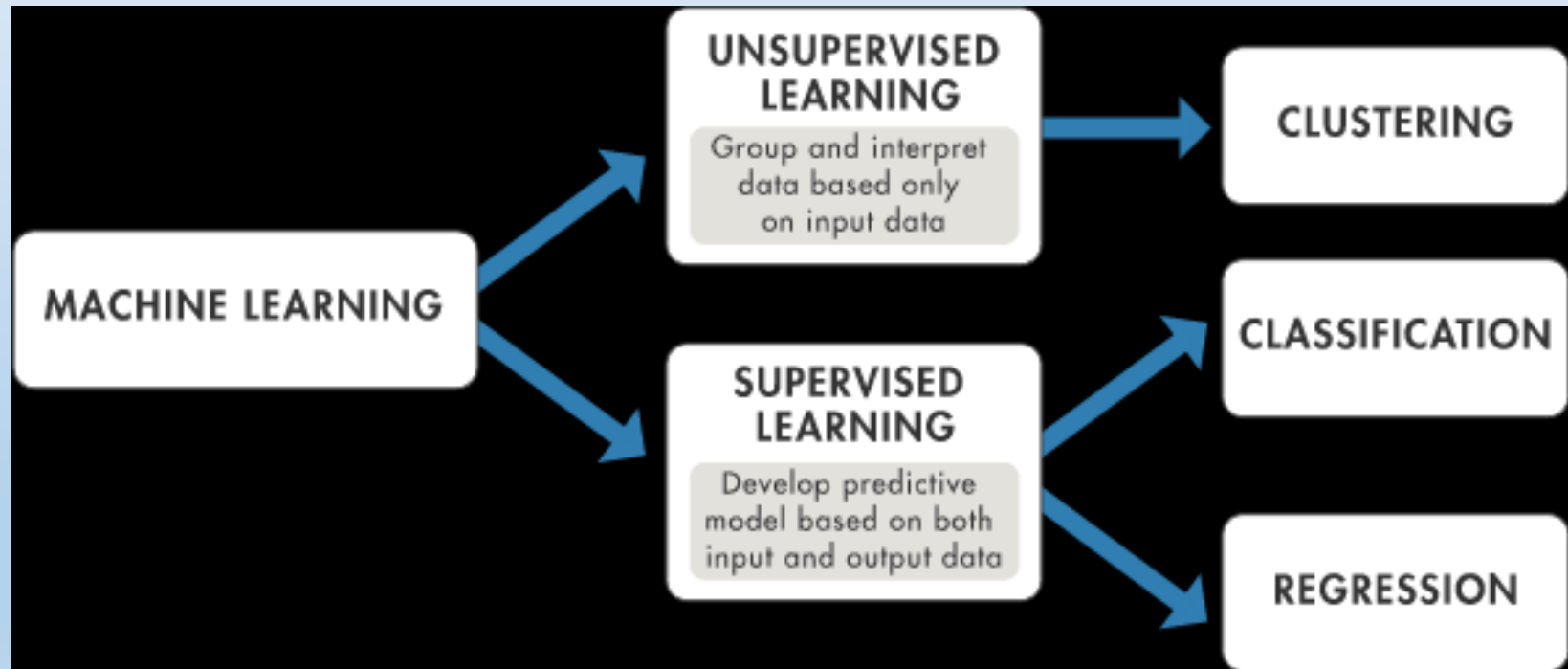


Definitions

- AI refers to systems that are intended to mimic human intelligence
- Machine Learning (ML) is a subset of AI
 - algorithms used to create models that improve over time by learning from data
- Deep Learning (DL) is a subset of AI
 - uses deep artificial neural networks to recognize complex patterns similarly to how the human brain processes information

Machine Learning Techniques (1)

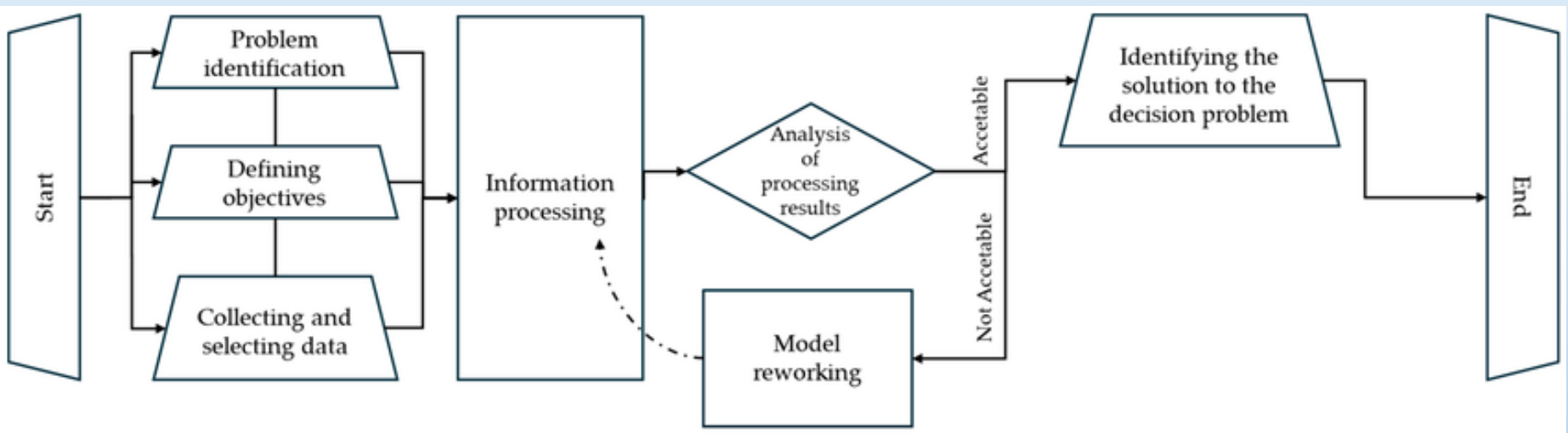
- <https://www.mathworks.com/discovery/machine-learning.html>



Machine Learning Techniques (2)

- **Unsupervised learning** a machine learning approach that uses algorithms to analyze and find patterns in unlabeled data. Data has no predefined categories or outcomes.
- **Supervised learning** a machine learning approach that trains algorithms using labeled datasets
 - **Classification techniques** - predict discrete responses (e.g., if an object was detected, if an e-mail is spam)
 - Support Vector Machine (SVM)
 - Decision Trees
 - Markov Decision Processes (MDP)
 - Item Response Theory (IRT)
 - Random forest
 - Naïve Bayes
 - Logistic regression
 - Neural networks
 - KNN
 - **Regression techniques** - predict continuous responses (e.g., changes in weather, parameters of a machine)
 - Regression linear model
 - Regression nonlinear model
 - Regularization
 - Neural networks
 - Adaptive neuro-fuzzy learning

Intelligent Decision-making System Flowchart



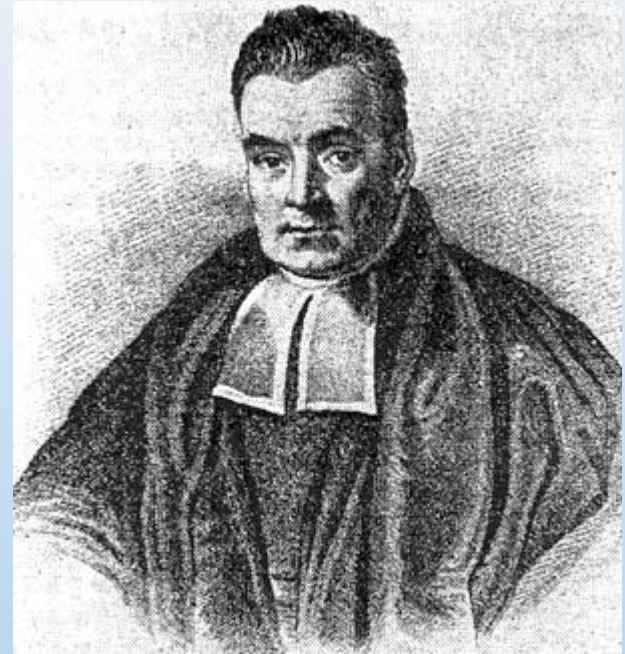
- https://www.researchgate.net/figure/Flow-chart-of-a-decision-making-process-AI-based-Source-authors-elaboration_fig1_385410511

User Modeling

- A field of HCI, AI, SE
- Concerned with creating of an internal representation of users to understand their behaviors, beliefs, goals, intention and needs
- The user interface should include “interface agents” or “intelligent agents” that can predict the user’s future actions based on:
 - the current state of the system
 - the user’s profile
 - previous actions
 - prior beliefs
- Therefore, user modeling uses a decision model that takes into consideration users’ behaviors, beliefs, goals, intention and needs
- The main question: How can we learn from such little evidence?

Bayesian Models History

- Reverend Thomas Bayes (1702 – 1761) mathematician and Presbyterian minister
- His theory on “inverse probability” was accepted by Laplace in 1781
- Bayes’s theorem was published posthumously
- Bayes argued that **the estimated probability for a future event is based on measurements of data, a model for these measurements, and a model for our prior beliefs about the system**



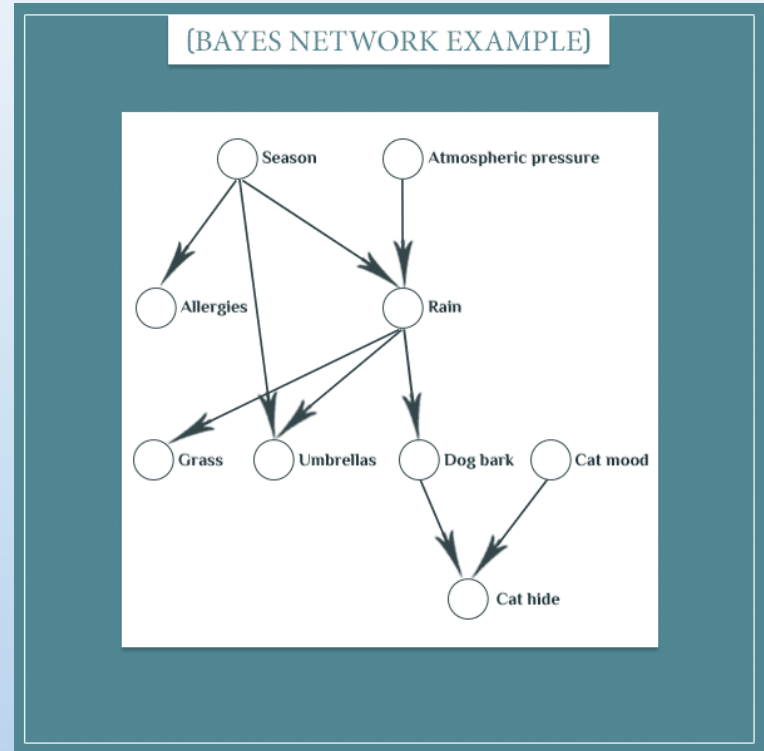
Bayesian Theorem

- The Bayesian Theorem uses “conditional probabilities”
- $P(H|X) = (P(X|H) * P(H)) / P(X)$
 - Where:
 - H is the hypothesis
 - X is the data
 - $P(H|X)$ is the posterior probability of H conditioned on X
 - $P(H)$ is the prior probability of H
 - $P(X|H)$ is the posterior probability of X conditioned on H
 - $P(X)$ is the prior probability of X (or normalized constant)
- $P(H|X) = (\text{likelihood} * \text{prior}) / \text{normalized constant}$

Bayesian Network (1)

- Represented by a graph
 - The nodes represent the random variables
 - The relationships between (variables) nodes are represented with directed edges (arrows) between pairs of nodes
 - A Bayesian network should be DAG (Directed Acyclic Graph) – no loops
- Includes conditional probability specifications
 - The conditional probability of each variable is given with respect to its parents in the DAG

Bayesian Network Example



- <https://www.probabilisticworld.com/bayesian-belief-networks-part-1/>
- <https://www.probabilisticworld.com/bayesian-belief-networks-part-2/#more-2391>

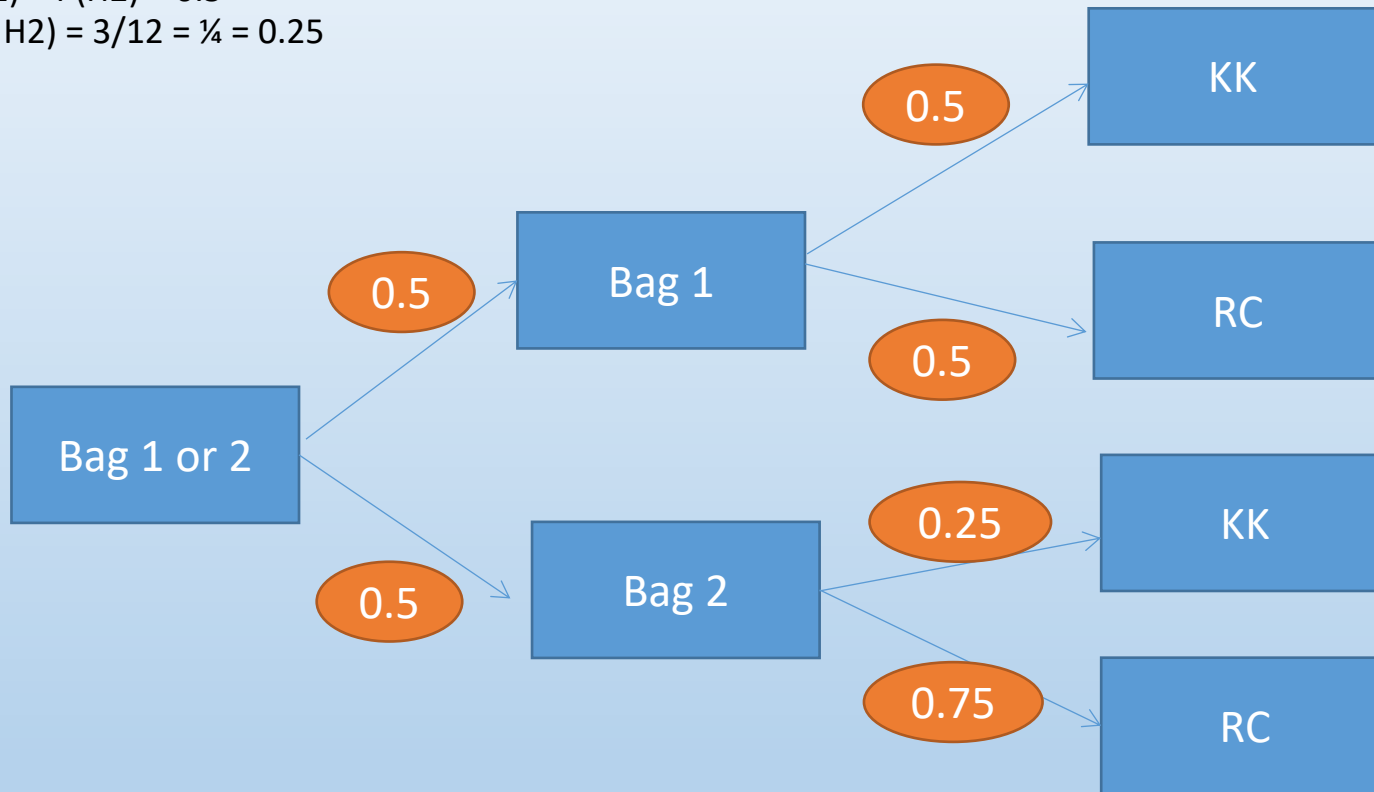
Example 1

- I have two bags each with 12 items. Bag 1 has 6 Kit-Kats and 6 Crispy cookies and bag 2 has 3 Kit-Kats and 9 Crispy cookies. I take one item from one of the two bags and it is a Kit-Kat. What is the probability that I took the candy from bag 2?
- Solution:
 - Hypothesis: H_1 = bag 1; H_2 = bag 2
 - Formula: $P(H_2|X) = (P(X|H_2) * P(H_2)) / P(X)$
 - Prior probability of a bag: $P(H_1) = P(H_2) = 0.5$
 - Posterior probability of a Kit-Kat candy conditioned on bag H_1
 - $P(X|H_1) = 6/12 = 1/2 = 0.5$
 - Posterior probability of a Kit-Kat candy conditioned on bag H_2
 - $P(X|H_2) = 3/12 = 1/4 = 0.25$
 - Prior probability of a Kit-Kat candy from any bag: $P(X) = P(H_1)*P(X|H_1) + P(H_2)*P(X|H_2) = 0.5*0.5 + 0.5*0.25 = 0.25 + 0.125 = 0.375$
 - $P(H_2|X) = (0.25 * 0.5) / 0.375 = 0.125/0.375 = 0.33$

Example 1 (2)

$$P(H1) = P(H2) = 0.5$$

$$P(X|H2) = 3/12 = \frac{1}{4} = 0.25$$



$$P(X) = P(H1)*P(X|H1) + P(H2)*P(X|H2) = 0.5*0.5 + 0.5*0.25 = 0.25 + 0.125 = 0.375$$

$$P(H2|X) = (0.25 * 0.5) / 0.375 = 0.125 / 0.375 = 0.33$$

Example 2

- A patient enter the doctor's office and complains that he is tired and has a bad cough
- The doctor considers three possibilities:
 - Flu
 - Lung cancer
 - Stomach flu
- What is the estimated probability for the patient to be sick with one of these disease?
- $P(H|X) = (\text{likelihood} * \text{prior}) / \text{normalized constant}$
- https://www.youtube.com/watch?v=0NGmrwu_BkY

Example 3

- <https://www.youtube.com/watch?v=pPTLK5hFGnQ>

Stochastic Systems

- The term "stochastic" means "pertaining to chance"
- A system is deterministic if it produces the same output for a given input
 - For example, software applications should be deterministic (this is a requirement)
- In the case of a stochastic system, disturbances can randomly produce different results on the same given input

Machine Learning (ML)

- ML algorithms are intended to identify patterns in used to make decisions and predictions
- ML techniques can be used to update the user models dynamically
- The Bayesian theorem is used in machine learning for classification.
- What is machine learning?
 - https://www.youtube.com/watch?v=f_uwKZIAeM0
- What is the difference between AI and ML?
 - <https://www.youtube.com/watch?v=J4Qsr93L1qs>
- Machine Learning Algorithms Overview
 - <https://www.youtube.com/watch?v=gglk08PNcBo>
- Naïve Bayes
 - <https://www.youtube.com/watch?v=O2L2Uv9pdDA>

Naïve Bayes

- Naive Bayes is a simple supervised ML algorithm that uses the Bayes' theorem
- It is used for text analysis
- It is called “Naïve” as it uses the assumption each input variable is independent
- Example: It will analyze the text as independent entities, e.g., :”I like COMP 4957” each word is independent
- It is used for classification

User Modeling Methods

- User modeling is used to identify and predict the users' **needs** based on a decision model that takes into consideration the users' **beliefs, intentions and goals**, an **existing model** and the **current data** collected during execution
- Bayesian models are used as **probabilistic models** for computing the likelihood of a certain event to happen

Can We Use Statistical and Probabilistic Models for User Modeling?

- Yes, we can
- Increased computational power
- Easy access to user data (i.e. repositories of data, logging activities are recorded)
- Conducting usability studies to understand our problem

Solutions

- Adaptive and adaptable interface design
 - is supposed to be the best outcome for users by keeping a user profile and predicting what the users' goals and needs might be
 - Adaptable interfaces: can be adapted by users
- Hybrid design - favored
- Adaptive interfaces are intelligent interfaces that adjust automatically to what

User Modeling Issues

- Complexity
- User modeling problems are still dominated by uncertainty
- More research is needed in areas like emotional needs (research was generally focused on cognition), culture, diversity, etc.